

## SAFETY DATA SHEET

Creation Date 22-Sep-2009

Revision Date 21-Apr-2022

Revision Number 5

### 1. Identification

**Product Name** N-Formylmorpholine

**Cat No. :** AC133800000; AC133800050; AC133801000; AC133805000

**CAS No** 4394-85-8  
**Synonyms** 4-Morpholinecarboxaldehyde

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Emergency Telephone Number**

For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Skin Sensitization

Category 1B

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

May cause an allergic skin reaction

**Precautionary Statements****Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray

Contaminated work clothing should not be allowed out of the workplace

Wear protective gloves

**Skin**

IF ON SKIN: Wash with plenty of soap and water

If skin irritation or rash occurs: Get medical advice/attention

Wash contaminated clothing before reuse

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

**3. Composition/Information on Ingredients**

| Component                  | CAS No    | Weight % |
|----------------------------|-----------|----------|
| 4-Morpholinecarboxaldehyde | 4394-85-8 | <= 100   |

**4. First-aid measures**

|                                            |                                                                                                                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                  |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                         |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                       |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                            |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                                                                                            |
| <b>Most important symptoms and effects</b> | May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                   |

**5. Fire-fighting measures**

|                                       |                                                                              |
|---------------------------------------|------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                                     |
| <b>Flash Point</b>                    | 113 °C / 235.4 °F                                                            |
| <b>Method -</b>                       | No information available                                                     |

**Autoignition Temperature** 345 °C / 653 °F

**Explosion Limits**

**Upper** 8.20%

**Lower** 1.20%

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up** Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

## 7. Handling and storage

**Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage.**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Incompatible Materials. Strong oxidizing agents.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.

**Engineering Measures**

None under normal use conditions.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                            |
|----------------------------------------|----------------------------|
| Physical State                         | Liquid                     |
| Appearance                             | Colorless to yellow        |
| Odor                                   | No information available   |
| Odor Threshold                         | No information available   |
| pH                                     | 10 - 250 g/l aq.sol        |
| Melting Point/Range                    | 20 - 23 °C / 68 - 73.4 °F  |
| Boiling Point/Range                    | 240 °C / 464 °F @ 760 mmHg |
| Flash Point                            | 113 °C / 235.4 °F          |
| Evaporation Rate                       | No information available   |
| Flammability (solid,gas)               | Not applicable             |
| Flammability or explosive limits       |                            |
| Upper                                  | 8.20%                      |
| Lower                                  | 1.20%                      |
| Vapor Pressure                         | 3.4 mbar @ 80 °C           |
| Vapor Density                          | No information available   |
| Specific Gravity                       | 1.140                      |
| Solubility                             | No information available   |
| Partition coefficient; n-octanol/water | No data available          |
| Autoignition Temperature               | 345 °C / 653 °F            |
| Decomposition Temperature              | No information available   |
| Viscosity                              | 6.79 cP at 30 °C           |
| Molecular Formula                      | C5 H9 N O2                 |
| Molecular Weight                       | 115.13                     |

## 10. Stability and reactivity

|                                  |                                                                                |
|----------------------------------|--------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                     |
| Stability                        | Stable under recommended storage conditions.                                   |
| Conditions to Avoid              | Incompatible products.                                                         |
| Incompatible Materials           | Strong oxidizing agents                                                        |
| Hazardous Decomposition Products | Nitrogen oxides (NOx), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ) |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                       |
| Hazardous Reactions              | None under normal processing.                                                  |

## 11. Toxicological information

### Acute Toxicity

**Product Information** Product does not present an acute toxicity hazard based on known or supplied information

### Component Information

| Component                  | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation               |
|----------------------------|---------------------------|-------------------------------|-------------------------------|
| 4-Morpholinecarboxaldehyde | LD50 = 6500 µL/kg ( Rat ) | LD50 > 18400 mg/kg ( Rabbit ) | LC50 > 5.319 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | May cause an allergic skin reaction                                                      |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component             | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------|-----------|------------|------------|------------|------------|------------|
| 4-Morpholinecarboxald | 4394-85-8 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                                   |                                                                                                                                                                                       |  |  |  |  |  |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| ehyde                                             |                                                                                                                                                                                       |  |  |  |  |  |
| <b>Mutagenic Effects</b>                          | No information available                                                                                                                                                              |  |  |  |  |  |
| <b>Reproductive Effects</b>                       | No information available.                                                                                                                                                             |  |  |  |  |  |
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                             |  |  |  |  |  |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                             |  |  |  |  |  |
| <b>STOT - single exposure</b>                     | None known                                                                                                                                                                            |  |  |  |  |  |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                            |  |  |  |  |  |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                              |  |  |  |  |  |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |  |  |  |  |  |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                              |  |  |  |  |  |
| <b>Other Adverse Effects</b>                      | .                                                                                                                                                                                     |  |  |  |  |  |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. .

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely                                               |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                             |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its water solubility. |

| Component                  | log Pow |
|----------------------------|---------|
| 4-Morpholinecarboxaldehyde | -1.2    |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

|                 |               |
|-----------------|---------------|
| <b>DOT</b>      | Not regulated |
| <b>TDG</b>      | Not regulated |
| <b>IATA</b>     | Not regulated |
| <b>IMDG/IMO</b> | Not regulated |

## 15. Regulatory information

### United States of America Inventory

| Component                  | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|----------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| 4-Morpholinecarboxaldehyde | 4394-85-8 | X    | ACTIVE                                        | PMN                         |

### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

PMN - Indicates a commenced PMN substance

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical** Not applicable

**Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

TSCA 12(b) - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                  | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|----------------------------|-----------|-----|------|-----------|-------|------|------|------|-------|----------|
| 4-Morpholinecarboxaldehyde | 4394-85-8 | X   | -    | 224-518-3 | X     | X    |      | X    | X     | 97-3-115 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations**

SARA 313

Not applicable

SARA 311/312 Hazard Categories

See section 2 for more information

CWA (Clean Water Act)

Not applicable

Clean Air Act

Not applicable

OSHA - Occupational Safety and  
Health Administration

Not applicable

CERCLA

Not applicable

California Proposition 65

This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know  
Regulations

Not applicable

**U.S. Department of Transportation**

Reportable Quantity (RQ):

N

DOT Marine Pollutant

N

DOT Severe Marine Pollutant

N

U.S. Department of Homeland  
Security

This product does not contain any DHS chemicals.

**Other International Regulations**

Mexico - Grade

No information available

Authorisation/Restrictions according to EU REACH

Not applicable

| Component                  | CAS No    | REACH (1907/2006) -<br>Annex XIV - Substances<br>Subject to Authorization | REACH (1907/2006) -<br>Annex XVII - Restrictions<br>on Certain Dangerous<br>Substances | REACH Regulation (EC<br>1907/2006) article 59 -<br>Candidate List of<br>Substances of Very High<br>Concern (SVHC) |
|----------------------------|-----------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 4-Morpholinecarboxaldehyde | 4394-85-8 | -                                                                         | -                                                                                      | -                                                                                                                 |

Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS No | OECD HPV | Persistent Organic<br>Pollutant | Ozone Depletion<br>Potential | Restriction of<br>Hazardous |
|-----------|--------|----------|---------------------------------|------------------------------|-----------------------------|
|-----------|--------|----------|---------------------------------|------------------------------|-----------------------------|

|                            |               |                                                                                                  |                                                                                                 |                                   |                                           |
|----------------------------|---------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------|
|                            |               |                                                                                                  |                                                                                                 |                                   | <b>Substances (RoHS)</b>                  |
| 4-Morpholinecarboxaldehyde | 4394-85-8     | Listed                                                                                           | Not applicable                                                                                  | Not applicable                    | Not applicable                            |
| <b>Component</b>           | <b>CAS No</b> | <b>Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification</b> | <b>Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements</b> | <b>Rotterdam Convention (PIC)</b> | <b>Basel Convention (Hazardous Waste)</b> |
| 4-Morpholinecarboxaldehyde | 4394-85-8     | Not applicable                                                                                   | Not applicable                                                                                  | Not applicable                    | Not applicable                            |

## 16. Other information

**Prepared By** Regulatory Affairs  
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**Creation Date** 22-Sep-2009  
**Revision Date** 21-Apr-2022  
**Print Date** 21-Apr-2022  
**Revision Summary** SDS sections updated.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**